SLIDE 1 · PROBLEM & MOTIVATION

When a Wind Turbine Goes Dark, How Much Energy Was Lost?

Turning missing blackout data into an auditable Business Interruption insurance claim.

1. The Grid Outage

- Cable damage or grid faults disconnect the wind farm from the grid.
- Turbines stop generating electricity and revenue drops to zero.

2. The Sensor Blind Spot

- Normally, we calculate lost energy using the turbine's own wind sensor.
- During a complete blackout, that wind sensor is offline too (missing NaN data).

3. Our Automated Solution

- Learn how neighbouring wind parks (Park O & Park H) relate to our turbine before the outage.
- Reconstruct the missing wind speeds and calculate lost MWh through the power curve.

Goal: Accurate, transparent loss claims that any insurance adjuster can verify | 72h Fleet Outage Loss: 283.9 MWh

SLIDE 2 · EXPLAINABLE ARCHITECTURE

Why Our Model Is Simple, Physical & 100% Explainable

Prioritizing explainability — one clean equation per turbine instead of a black box.

One Readable Equation

- Uses Bayesian Ridge Regression.
- Every turbine gets a single linear formula you can verify on a calculator.
- Zero hidden layers or opaque decision rules.

Why Aggregate into Parks?

- Turbines in the same park (O1–O2 or H1–H5) feel >96% identical wind.
- Separate turbine weights fight each other and create fake negative numbers.
- Park medians give clean ~40% / ~40% weights and survive single-sensor dropouts.

Better Than Other Models

- vs. LightGBM (450 Trees): Black box that fails seasonal shifts (-20% error) and cannot be audited.
- vs. Raw OLS Regression: Suffers from unphysical negative weights.
- Explainability is key for insurance sign-off.

Formula Structure: $v_{Target} = w0 + w_O \cdot Speed(Park\ O) + w_H \cdot Speed(Park\ H) + Direction\ (u, v)$ | 100% Transparent

SLIDE 3 · NATIVE UNCERTAINTY QUANTIFICATION

Native Bayesian Uncertainty: Two Bands, Zero Crossing

Uncertainty comes directly from Bayesian probability math — not ad-hoc quantile trees.

Why Native Bayesian Beats Quantile Models

- Quantile Models (LightGBM): Train 3 separate black boxes (P10, P50, P90). Because they don't share math, curves can cross (P10 > P50).
- Native Bayesian Ridge: Outputs a complete Gaussian bell curve (mean μ & spread σ) at every 10-minute step.
- Guaranteed Consistency: Any percentile (50%–99% slider in our app) is smooth and mathematically cannot cross.

Two Uncertainties Separated Automatically

- 1. 95% Coefficient CI (Dark Inner Band — ±2.0 MWh): 'How sure are we about our formula?' Pinpointed tightly by 12,400+ training intervals.
- 2. 80% Predictive Band P10–P90 (Light Outer Band): 'How gusty is a single 10-minute step?'
- Over a 3-day outage, random 10-minute wind gusts cancel out — so true energy stays close to P50.

T1 Outage Example: Central P50 = 75.3 MWh · 95% Formula CI = 73.3–77.4 MWh · 80% Gust Band = 26.8–154.8 MWh | Calibrated at 82.7%

SLIDE 4 · CROSS-VALIDATED ACCURACY

Robust Accuracy via 5-Fold Cross-Validation

Tested across all 86 operational days (12,421 intervals) without permanently withholding training data.

Turbine T1 (5-Fold CV)

- Wind Speed MAE: 0.990 m/s
- Wind Speed RMSE: 1.338 m/s
- Power MAE: 492.6 kW
- 80% Band Coverage: 82.4%
- 5-Fold Energy Error: -12.78%
- 72h Outage Loss: 75.32 MWh

Turbine T2 (5-Fold CV)

- Wind Speed MAE: 0.880 m/s (Best)
- Wind Speed RMSE: 1.182 m/s (Best)
- Power MAE: 453.4 kW
- 80% Band Coverage: 82.8%
- 5-Fold Energy Error: -8.77%
- 72h Outage Loss: 117.16 MWh

Turbine T3 (5-Fold CV)

- Wind Speed MAE: 0.893 m/s
- Wind Speed RMSE: 1.194 m/s
- Power MAE: 440.2 kW (Best)
- 80% Band Coverage: 83.0%
- 5-Fold Energy Error: -11.35%
- 72h Outage Loss: 91.46 MWh

Why Background 5-Fold CV? Tests every time block out-of-sample while the final model trains on 100% of history | Zero Data Wasted

SLIDE 5 · PHYSICS & BONUS FEATURES

Physics Addons & All 3 Bonus Challenges

Going beyond wind speed with atmospheric physics, smart filtering, and instant claim emails.

1. Direction & Air Density Physics

- Wind Vectors (u, v): Fixes the 359° ↔ 1° compass jump and captures hill & wake sheltering.
- IEC 61400-12-1 Air Density: Uses live Temperature & Pressure from Park H (97.9% regional match) to adjust blade push for thin warm vs. dense cold air.

2. Auto Reference Selection

- Automatically scans all uploaded reference files (O1–O2, H1–H5).
- Detects and removes any reference turbine that is offline during the outage.
- Ranks references by correlation and shared training days.

3. One-Click Insurance Claim

- Generates a clean claim notice with only relevant data: outage times, reference turbines, Estimated MWh, and 5-Fold CV proof.
- Includes an 'Open in Default Mail App' button that launches your email client pre-filled.

Live Demo: Interactive Streamlit App with plain-English '?' guides, manual calendars & 10-min certainty explorer | BayWa r.e. Ready